

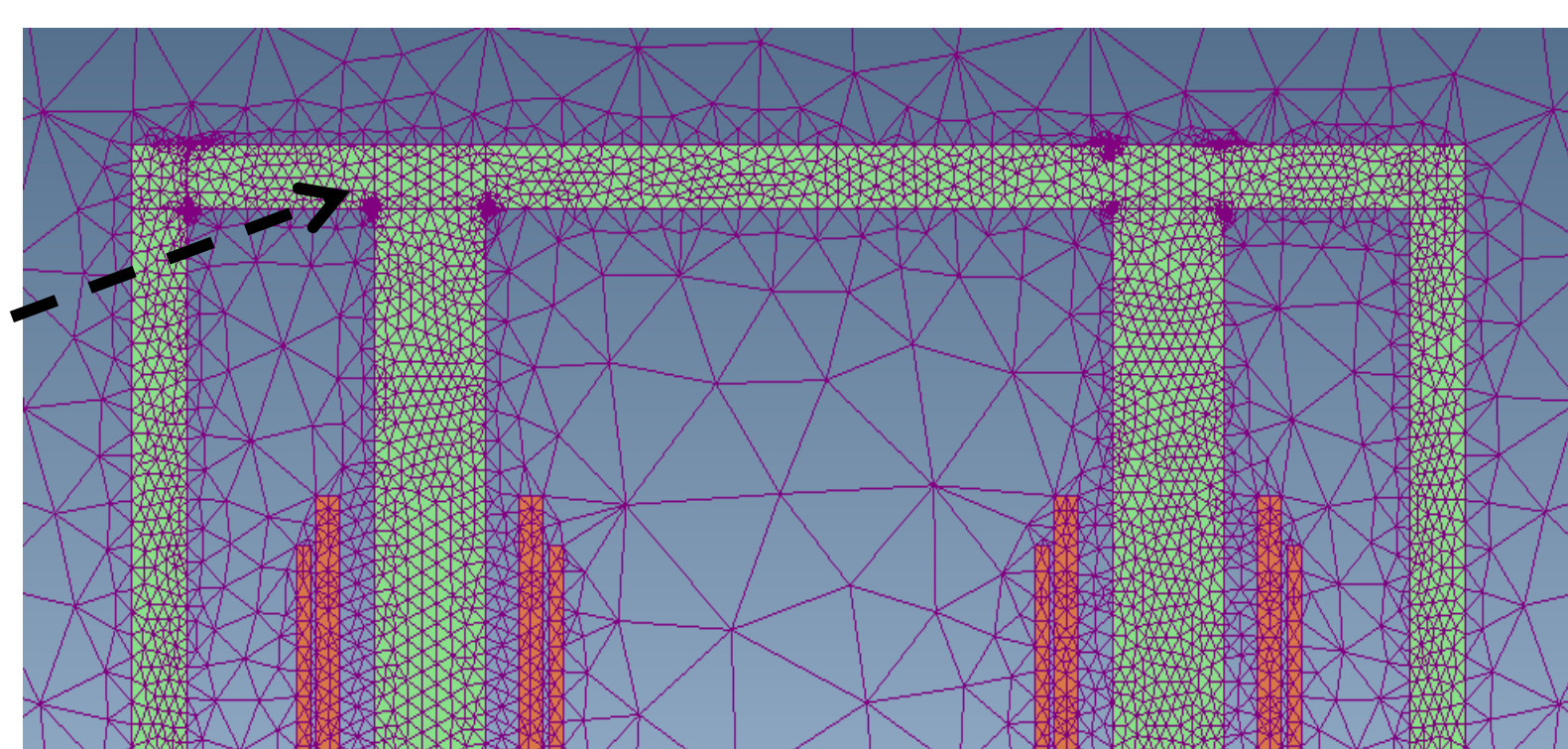
Measurements and finite element model of transformer core joints with dc and ac excitation

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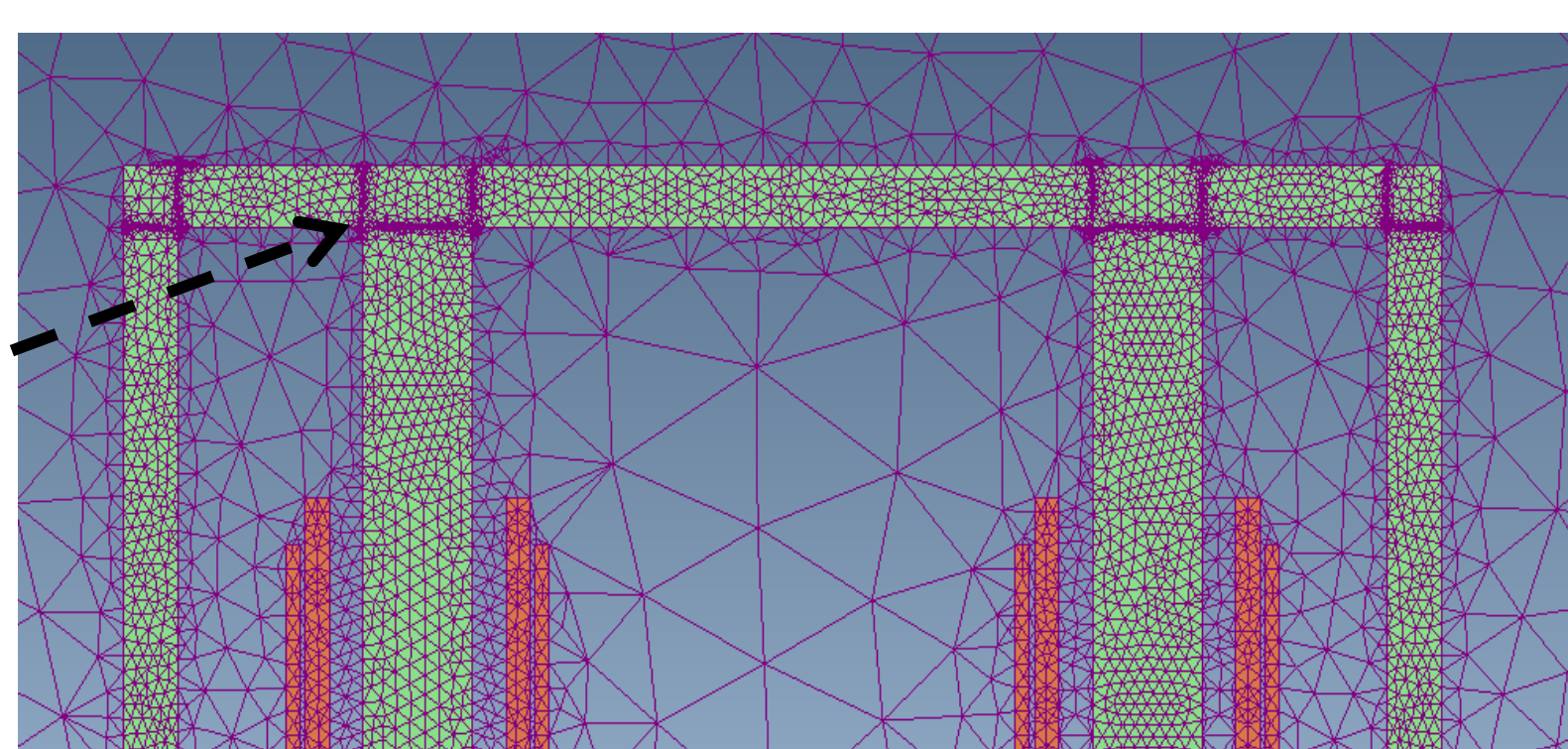
Background

- GIC enter PTNs through Power Transformers as quasi-dc currents ($f < 1$ Hz)
- Half-cycle saturation = many unwanted conditions in PTNs
- Stray flux = hot spots, insulation degradation and gassing
- Factory Testing with dc not practically feasible – excessive draw of Mvars
- GIC withstand design models are needed but validation with measurement data is the challenge

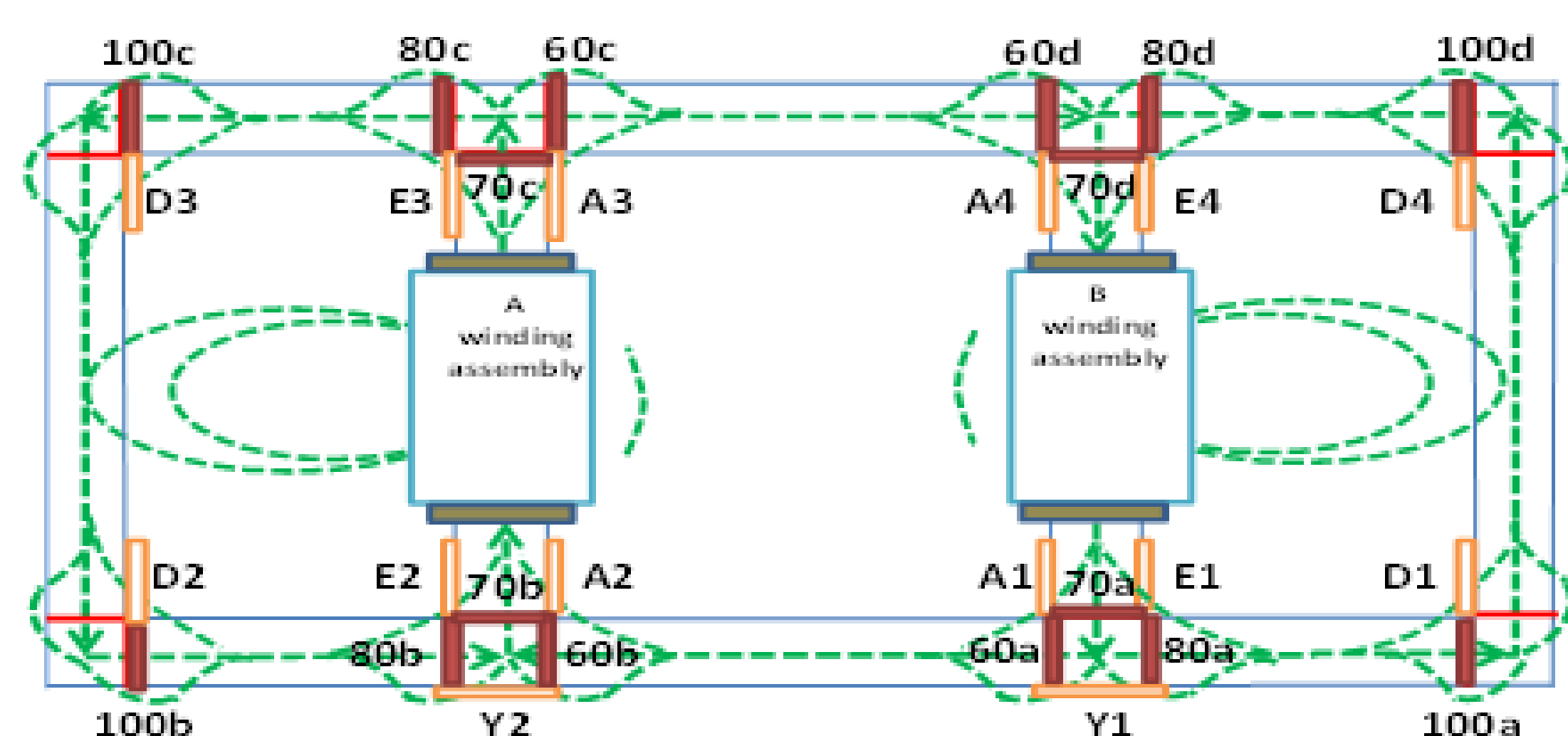
FEM S model with solid joints



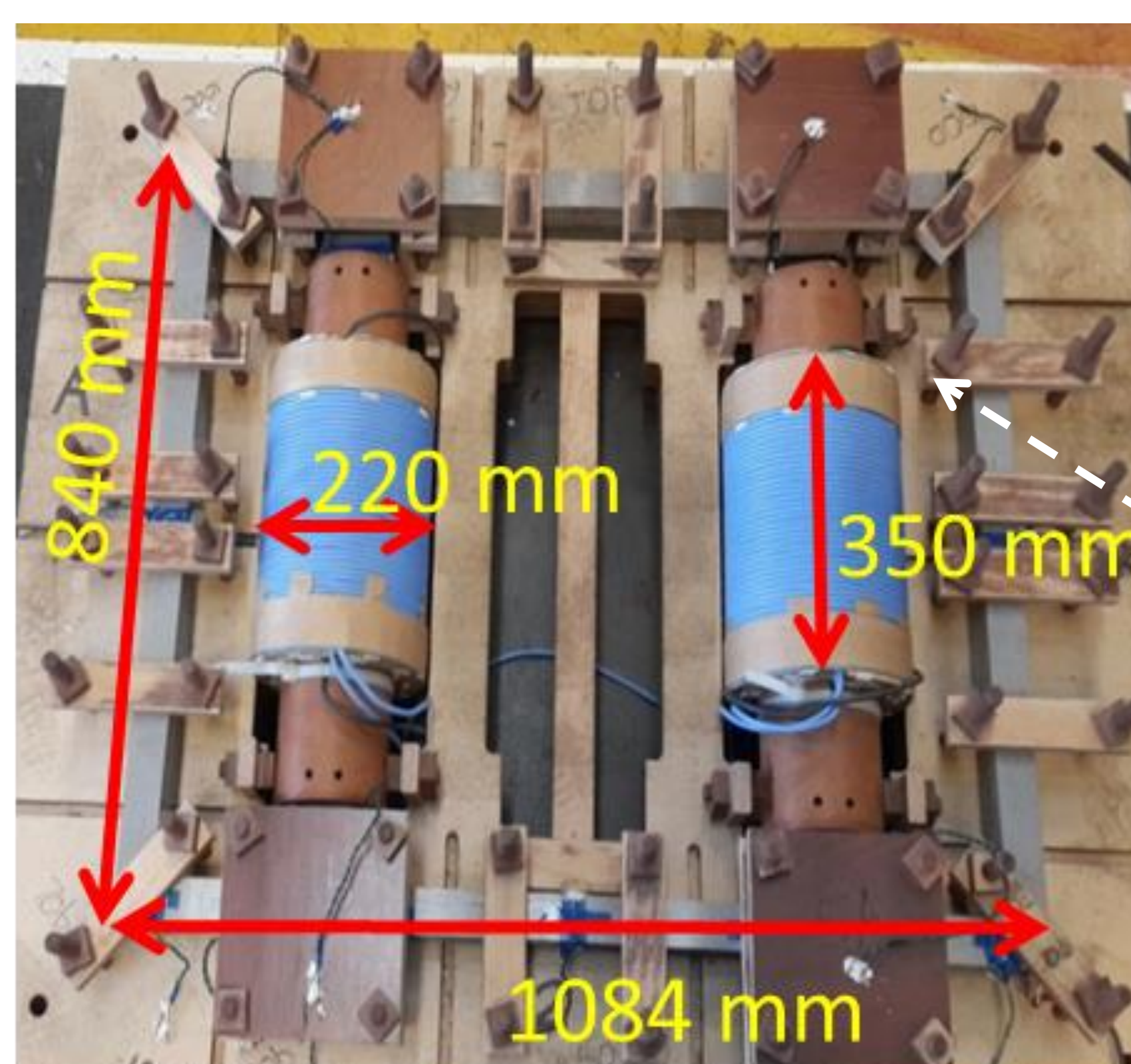
FEM B-AG model with “equivalent air gaps” at the joints



Approach

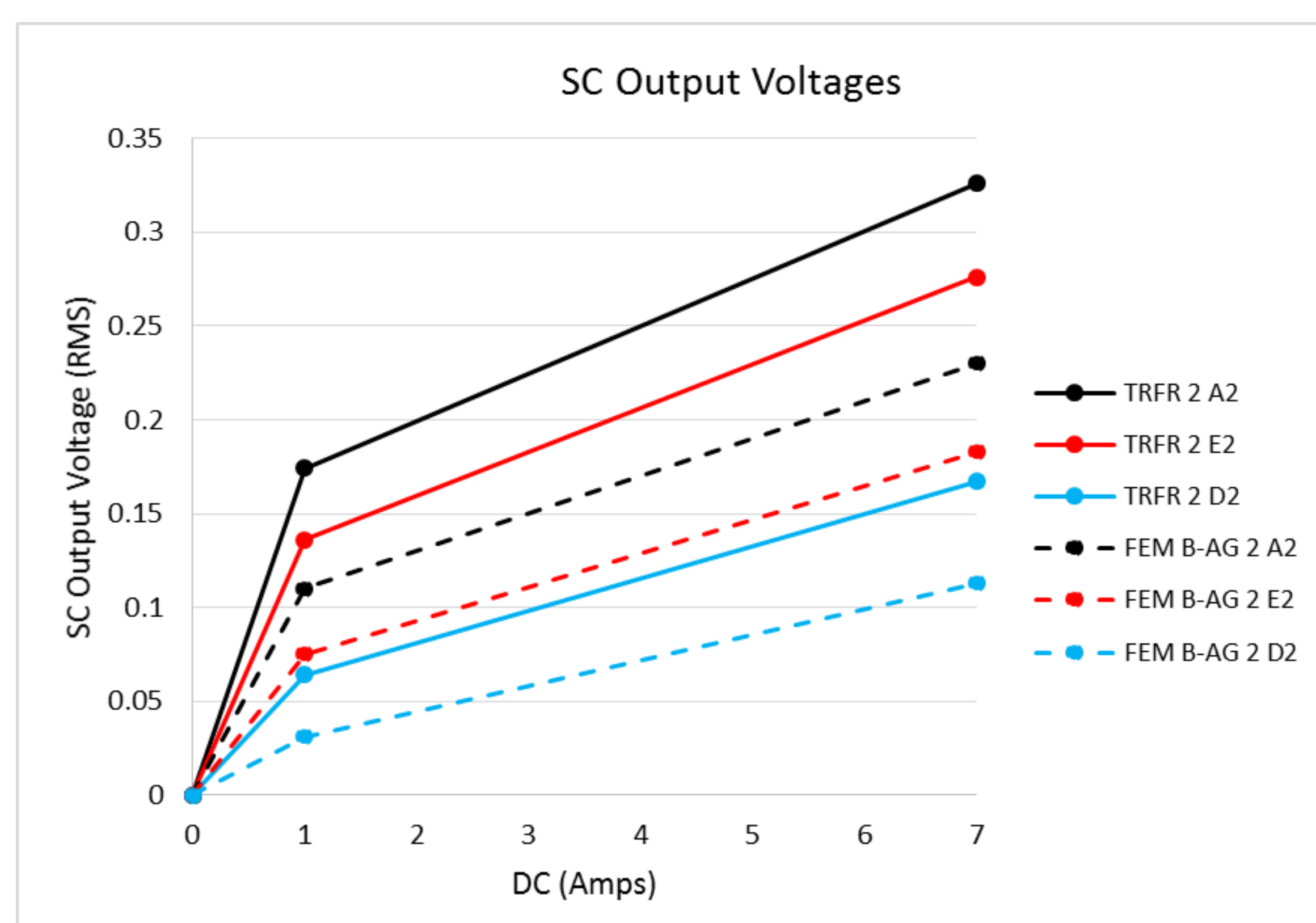


- Laboratory flux measurements with 3X1p-4L transformers [1]
- Nominal ac applied + various dc levels with search coil measurements
- FEM models with different core joint detail investigated



One single-phase four limb laboratory transformer - 110/206 V, 4.4 kVA.

Results



Measured (TRFR 2) and FEM B-AG air SC output voltages with simultaneous ac and dc excitation

Conclusions

- FEM models calibrated with ac measurement data
- No search coil outputs for solid joint FEM model with ac + dc
- “Equivalent air gap” FEM models give results closer to those measured with ac + dc together
- More detailed numerical modelling needed for more accuracy - *anisotropy*

[1] L. D. Borrill, H. K. Chisepo and C.T. Gaunt, *Flux measurements with ac and dc components of current present show transformer equivalent circuit models need core joint details*, In Proc. IEEE POWERCON, Wollongong, Australia, Sept. 2016



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